Big Data, Data Science and Data Analytics

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Big Data

What is “Big Data”?

A buzz word with fuzzy boundaries:

- “Massive amounts of diverse, unstructured data produced by high-performance applications”
- “Data too large & complex to be effectively handled by standard database technology currently found in most organisations”

But what is “Big Data”, really?

The 5Vs of Big Data

Big Data is data affected by the 5Vs property:

- **Volume**: large amounts of data
- **Variety**: data comes in many different forms from diverse sources
- **Velocity**: the content is changing very quickly

Initially only 3V … but now there are other 2:

- **Value**: data alone is not enough; how can value be derived from it?
- **Veracity**: can we trust the data? how accurate is it?

Volume

This is easy …

- YouTube: 4 billion views a day, one hour of video uploaded every second
- Facebook: 483 million daily active users (Dec. 2011); 300 Petabytes of data
- Google: >1 billion searches per day (March 2011)
- Google processed 100 Terabytes of data per day in 2004 and 20 Petabytes data/day in 2008
- Internet Archive: contains 2 Petabytes of data, grows 20 Terabytes per month (2011)

Velocity

- **Batch processing**: running a series of computer programs without human intervention

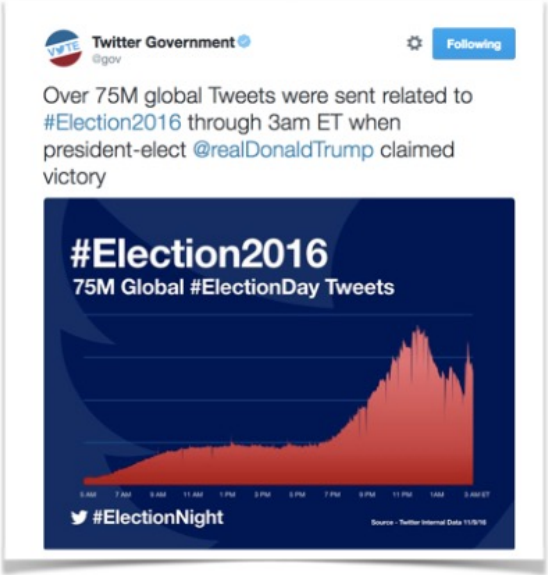
- **Near real-time**: brief delay between the data becoming available and it being processed

- **Real-time**: guaranteed almost *immediate* responses between the data becoming available and it being processed

Volume and Velocity, together

Things become more interesting when these elements are combined

- **>500 million** tweets a day
- **>300 million** active users / day
- On average >6000 tweets a second
- Peaks of >100,000 tweets a second
  - Super Bowl
  - US election (**75M** tweets in 24h)
  - New Year's Eve
  - Football World Cup (**672M** tweets in total #WorldCup)



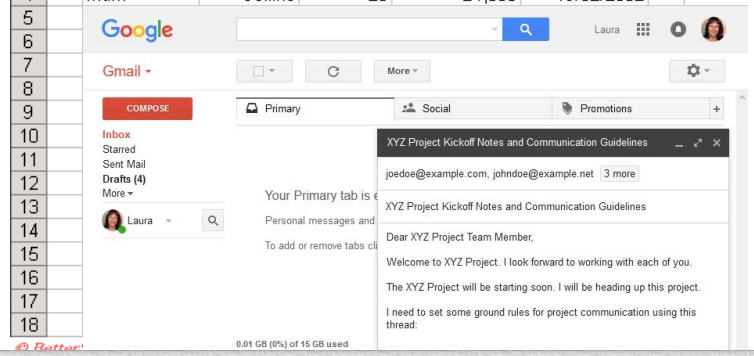
- Messages are instantly accessible for **search**
- Messages are used in **post-hoc** analyses to gather insights

Variety

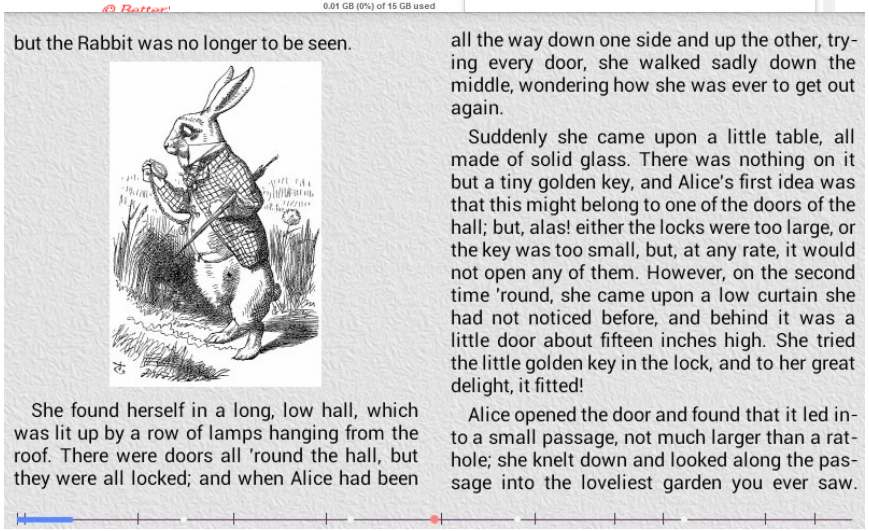
Structured data

First Name	Last Name	Age	Salary	Date
John	Smith	19	18,000	14/04/2003
Mark	Collins	20	24,000	19/02/2002

Semi-structured data



Unstructured data



but the Rabbit was no longer to be seen.

She found herself in a long, low hall, which was lit up by a row of lamps hanging from the roof. There were doors all 'round the hall, but they were all locked; and when Alice had been all the way down one side and up the other, trying every door, she walked sadly down the middle, wondering how she was ever to get out again.

Suddenly she came upon a little table, all made of solid glass. There was nothing on it but a tiny golden key, and Alice's first idea was that this might belong to one of the doors of the hall; but, alas! either the locks were too large, or the key was too small, but, at any rate, it would not open any of them. However, on the second time 'round, she came upon a low curtain she had not noticed before, and behind it was a little door about fifteen inches high. She tried the little golden key in the lock, and to her great delight, it fitted!

Alice opened the door and found that it led into a small passage, not much larger than a rat-hole; she knelt down and looked along the passage into the loveliest garden you ever saw.

Velocity and Variety, together

Task: Given a person's location anywhere in the world, list five restaurants in his immediate neighborhood that he might like to go to

Available data:

- World map, list of all restaurants in the world (opening hours, GPS coordinates, menu, special offers)
- Reviews/ratings
- Social media stream(s)
- User taste

Data is continuously *changing* (e.g., restaurants close, new ones open, data formats change, etc.)

Value

Think of the value of accurate *predictions of natural disasters* and diseases …

… or better responses to *disaster recovery*

- Timely & effective decisions
- Provide resources where they need the most

… or extensive disease/genomics databases to enable *biomedical discoveries*

… or accurate models to support *forecasting* of ecosystem developments

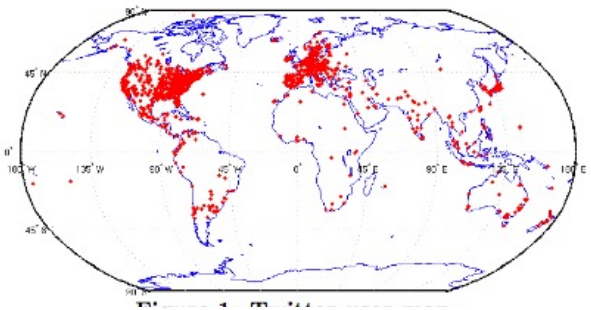
Value, Veracity and Velocity together

Earthquake Warnings

Social sensors: users (humans) that use Twitter, Facebook, Instagram, i.e., portals with *real-time* posting abilities

Challenges: how to detect when a tweet is about an *actual earthquake*, *which* earthquake it is, and *where* is its center

Value, Veracity and Velocity …



Twitter-based

Date	Magnitude	Location	Time	E-mail sent time	#tweets within 10 min	Announce of JMA
Aug. 18	4.5	Tochigi	6:58:55	7:00:30	35	07:08
Aug. 18	3.1	Suruga-wan	19:22:48	19:23:14	17	19:28
Aug. 21	4.1	Chiba	8:51:16	8:51:35	52	8:56
Aug. 25	4.3	Uraga-oki	2:22:49	2:23:21	23	02:27
Aug. 25	3.5	Fukushima	22:21:16	22:22:29	13	22:26
Aug. 27	3.9	Wakayama	17:47:30	17:48:11	16	17:53
Aug. 27	2.8	Suruga-wan	20:26:23	20:26:45	14	20:31
Aug. 31	4.5	Fukushima	00:45:54	00:46:24	32	00:51
Sep. 2	3.3	Suruga-wan	13:04:45	13:05:04	18	13:10
Sep. 2	3.6	Bungo-suido	17:37:53	17:38:27	3	17:43

earthquake 40 traditional warning system

Sakaki et al, 2010

In summary

Big data refers to a "*humongous*" amount of data that has the 5V properties, but …

…*what does this have to do with DA projects?*

DA projects are meant to give you *some first and small first hand experience* of what it means to *deal with Big Data with the limited resources you have available*

This involves Data Science and Data Analytics

Data Science

Data Science is concerned with mining large amounts of structured and unstructured data to identify patterns

A data scientist:

- Predicts the future based on past patterns
- Explores and examines data from multiple disconnected sources
- Develops new analytical methods and machine learning models

Data Analytics

Data Analytics processes and performs data analysis to discover how data can be used to draw conclusions and solve problems

A data analyst:

- Acquires, processes, and summarises data
- Packages data for insights
- Designs and creates reports using various reporting tools
- Tries to draw some initial conclusions from the data (but leaves deeper conclusions to domain experts)

Remember: this course is about Data Analytics

Questions

General Instructions for DA Projects

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General comments

There are *two project assignments* during the course; this is the first one

- The first assignment covers the material presented in the first part of the course

In this assignment there are 21 projects

- One for each pair of students
- Projects are also assigned randomly!
- If your project's companion drops the course or drops the project within the next 2 weeks, please contact the TA for project reassignment

What you are supposed to do

For each of the project you must:

1. Get the data of your project (see links provided)
2. Analyse the data (use standard descriptive statistics)
3. Make some hypothesis based on the characteristics of the data
4. Test the hypothesis, using some of the techniques learned from the theoretical part of the course
5. Draw some interesting conclusions

Examples of the process will be presented in the tutorial sessions

Example: Prediction

- Datasets: e.g. wine quality, intrusion, insurance, …
- Problem: you have values of some variables, predict the values of the missing ones
- Tasks: exploratory study, pre-process data, visualisation
- Hypothesis: build a simple prediction model
  - With the assumption that the data you have is a small sample of a much larger set
- Test: quality of your prediction

Example

Data Analytics

Academic Year 2019-20

Course Assignment N. 02: Predicting Employees Salary

Prof. Fabio Crestani

TAs: Esteban A. Ríssola, Manajit Chakraborty

For this assignment you will work individually to carry out simple tasks of data analysis given a specific dataset. The goal of this assignment is to use Python and complementary libraries on a given dataset in order to *explore* and *analyze* the given data and *draw conclusions*.

Description

The data lists demographic characteristics of people.

Your goal is to predict the income based on the provided features (or a set of them). Your tasks are to:

- Explore and describe the data (*i.e.*, standard descriptive statistics, visualize the variables with different graphs, draw distributions and histograms of variables, are there outliers? Any interesting observation? Any correlations? Etc.)
- Pre-process the data (*i.e.*, handle and fill unknowns if there are any, etc.)
- Build one prediction model to predict the income of the test samples (consider income as unknown – remove it first)
  - The real value of the variable income is given to be able to evaluate your result
- Evaluate and compare the performance of the model.

Submission procedure and evaluation

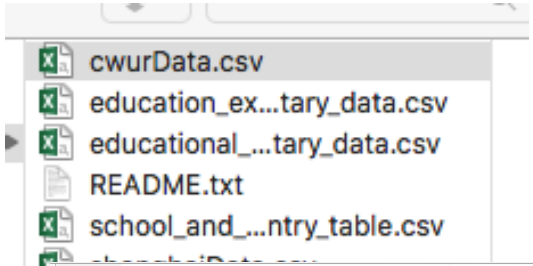
You should produce a report of your work and its evaluation along with the source code. It will be a concise explanation of how you tackled the different tasks, the reasons of your choices, successive conclusions, graphs you produced, results of the decisions and their accuracy, *etc.*

Use Jupyter Notebook to produce results of the commands in a single .ipynb file. For more information check: https://jupyter.org/documentation

The report (max 5 pages) and the code of the project need to be submitted via iCorsi.

Please, upload all the required items in a single file and name it following the structure: **noProject_FirstnameLastname.[zip|tar.gz|7z]**. For instance, 05_EstebanRissola.tar.gz

The dataset regarding this project can be downloaded from: http://ir.inf.usi.ch/da-datasets/



Data is provided in txt or cvs format

A	B	C	D	E	F	G	H	I	J	K	L	M	N
world_rank	institution	country	national_ran	quality_of_e	alumni_emp	quality_of_fa	publications	influence	citations	broad_impac	patents	score	year
1	Harvard Univ	USA	1	7	9	1	1	1	1		5	100	2012
2	Massachusel	USA	2	9	17	3	12	4	4		1	91.67	2012
3	Stanford Uni	USA	3	17	11	5	4	2	2		15	89.5	2012
4	University of	United Kingd	1	10	24	4	16	16	11		50	86.17	2012
5	California Ins	USA	4	2	29	7	37	22	22		18	85.21	2012
6	Princeton Un	USA	5	8	14	2	53	33	26		101	82.5	2012
7	University of	United Kingd	2	13	28	9	15	13	19		26	82.34	2012
8	Yale Univers	USA	6	14	31	12	14	6	15		66	79.14	2012
9	Columbia Un	USA	7	23	21	10	13	12	14		5	78.86	2012
10	University of	USA	8	16	52	6	6	5	3		16	78.55	2012
11	University of	USA	9	15	26	8	34	20	28		101	73.82	2012
12	Cornell Univ	USA	10	21	42	14	22	21	16		10	73.69	2012
13	University of	USA	11	31	16	24	9	10	8		9	73.64	2012
14	University of	Japan	1	32	19	31	8	19	23		3	69.49	2012
15	Johns Hopkin	USA	12	34	77	20	11	9	9		7	66.94	2012
16	Swiss Federa	Switzerland	1	26	66	11	40	51	44		34	66.69	2012
17	Kyoto Univer	Japan	2	42	38	19	25	36	43		23	65.76	2012
18	Weizmann In	Israel	1	4	101	22	101	67	101		29	65.09	2012
19	University of	USA	13	62	59	23	3	11	6		13	64.05	2012
20	University of	USA	14	61	101	15	10	8	10		22	63.11	2012
21	Rockefeller U	USA	15	1	101	16	101	28	96		101	61.74	2012
22	Hebrew Univ	Israel	2	24	93	13	101	91	101		28	60.76	2012
23	New York Ur	USA	16	89	75	17	42	24	34		62	60.55	2012
24	University of	USA	17	101	101	21	19	3	13		33	59.7	2012
25	University of	USA	18	64	63	33	17	30	21		21	59.66	2012
26	University of	USA	19	82	101	18	35	71	39		44	59	2012
27	Duke Univers	USA	20	65	43	55	20	15	12		20	58.37	2012
28	Imperial Coll	United Kingd	3	84	73	35	26	26	29		41	57.53	2012
29	University of	USA	21	19	101	32	101	43	84		101	56.43	2012
30	University of	USA	22	101	78	27	41	47	40		57	56.18	2012
31	University Co	United Kingd	4	35	101	45	27	23	33		86	55.21	2012
32	Osaka Unive	Japan	3	77	101	44	39	44	51		11	54.43	2012
33	Northwester	USA	23	101	32	101	24	25	20		35	54.4	2012
34	University of	USA	24	68	60	101	2	17	7		8	53.72	2012
35	University of	Canada	1	101	101	34	7	14	18		101	53.43	2012
36	University of	USA	25	101	86	56	31	29	31		29	53.09	2012

TA sessions

The TA will present examples of completed assignments using different techniques during their sessions

- See calendar for the dates

TAs are the first point of call for questions regarding the assignments

Submissions

You must submit:

1. The Python Jupyter notebook of your project

2. A report of your work (max 5 pages) in which you must explain:
   - how you tackled the tasks
   - the reasons of your decisions
   - the successive conclusions you made
   - any graphs you produced
   - results of the evaluation

Assignment deadlines

Normally you have about 6 weeks to carry out a project

- Project 1: 12 March -> 17 April
- Project 2: 18 April -> 23 May

Do not wait for the last days to start the work!

First project submission deadline is midnight April 17th

- Submission via iCorsi
- Do not miss the deadline

Guidelines for assignment completion

Assigment might be completed at different levels

- Please follow the completion guidelines posted on iCorsi
- Specifically, in connections with the report …

Different level of completion will determine different project marks!

The assignment submission consists of two deliverables, namely, the code (Jupyter Notebook) and the report (in pdf). While preparing the documents, we would recommend that you follow the following guidelines.

Code

1. **Executability**: The first and foremost requirement is that the code should execute without any errors for each and every cell.
2. **Readability**: As a good practice, try and provide comments explaining the purpose of code sections (which are non-trivial). Remember, we all have different coding styles, so we should be able to read and understand your code without hassle. E.g., assign representative names to variables (and not just x and y). It is advisable to use Markdown facility in the notebook to distinguish different sections.
3. **Reusability**: You are free to use any piece of online code, as long as you prove that you have a good understanding of its purpose and execution.
4. **Path Convention**: Remember to use relative paths instead of absolute ones, so that we can readily access the data from your code.

Report

1. The report is **NOT** a Readme document. You need to have a proper structure, including sections like (but not limited to): Introduction, Data Analysis and Experiments, Results, and Conclusions. The report (pdf) can be generated using LaTeX or any text editor.
2. The report should **NOT** be an exported (pdf) version of the Jupyter Notebook. Not everything that you present in the notebook needs to be included in the report. Be selective. *E.g.*, in the Data Analysis, you may provide only the most distinctive observations which support your hypothesis and showcases your comprehension of the task. Remember, there is a strict page limit.
3. The presentation of the report is an integral part of the evaluation. So, remember to name and refer each section, figure, and tables within the text properly.
4. In case, you need to use additional packages (not shown in class), provide the source URL, for instance, as a footnote in the text.

Your assignment will be evaluated on the following parameters:

1. **Executability**: If the code runs without errors or not.
2. **Data Preparation and Analysis**: How rigorous it is.
3. **Experimental Setup and Model**: The models you use, the motivation behind choosing a particular method/technique/algorithm.
4. **Report**: How well you follow the above-mentioned guidelines and aesthetics.

Project assignment 1

Please indicate the name of your preferred project partners in:

https://docs.google.com/spreadsheets/d/1VHIK2MToVBbIPVyEGX-0YXCw_KWT79i6wfDzLrHpStw/edit?usp=sharing

by March 9th, midnight (project 1) and April 14th midnight (project 2)

Projects for assignment 1 will be made available March 12th.

Questions

For any other question please contact the TAs